Shivam Patel

Data Scientist | AI/ML Engineer | NLP and LLM Applications

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Summary

Data Scientist and AI/ML engineer with experience building NLP pipelines, predictive models, LLM applications, and explainable AI workflows. Skilled in Python, model evaluation, feature engineering, data visualization, and deploying applied ML systems with AWS and Docker.

Education

M.S. Computer Science - San Diego State University, San Diego, CA *CGPA: 3.95/4 | Expected May 2027*
B.Tech Computer Science and Engineering - Indus University, Ahmedabad, India *CGPA: 9.73/10 | May 2025*

Experience

Instructional Student Assistant (ISA), Computer Networks| *San Diego State University* *Aug 2026 - Present*

- Supported CN instruction by assisting students with course concepts, grading assignments and taking tests.

Data Science Intern | *Embrizon Technologies* *Oct 2024 - Dec 2024*

- **Financial Sentiment Analysis:** Processed **80K+** financial tweets across **10** stocks and automated training workflows, reducing retraining time by **40%**.
- **Smart Healthcare Assistant:** Integrated four datasets containing approximately **5K** records into a TF-IDF and Linear SVC workflow; added XAI and severity scoring, reducing manual review time by **60%.**

Machine Learning Intern | *LogicSpark Technology* *Jun 2024 - Aug 2024*

- Integrated a **Random Forest** model with **hotel location** and **booking-period data** to improve room-price prediction and support data-driven rate adjustments.
- Automated daily margin analysis from partner booking requests, enabling dynamic pricing adjustments.

Teaching Assistant | *Indus University* *Jan 2023 - Dec 2023*

- Supported students in ML, Data Science and Calculus with Python, data processing and problem-solving guidance.

Projects

PDF Chatbot with LLM ↗ | *Python, Flask, LangChain, OpenAI API, Docker, AWS*

- Developed and deployed an AWS-hosted LLM application for question answering over PDFs up to **200 MB**.
- Evaluated **80** questions across 25 PDFs, achieving **100%** answer **accuracy** and **100%** relevant-document retrieval.
- Built a Flask backend for **10+ concurrent sessions** with Docker/CI/CD and **99.9% uptime**.

NYC Collision Analysis ↗ | *Python, Pandas, NumPy, Matplotlib, Seaborn*

- Analyzed **238K** collision records to identify injury patterns and driver-inattention trends.
- Benchmarked three classifiers with logistic regression achieving **0.60** F1, **80%** recall, and **0.66** ROC-AUC.

Global Volcano Activity Analysis and Modeling ↗ | *Python, NumPy, Scikit-learn, XGBoost, GeoPandas, Plotly*

- Analyzed **10,000+** eruptions across **12,000 years**; engineered features for 3-class VEI classification.
- Tuned XGBoost with SMOTE, achieving **85.5%** accuracy and **0.795** macro-F1.

Employee Management System ↗ | *React, Express.js, MongoDB, Mongoose, Webpack*

- Built a full-stack web application interface for creating, viewing, editing, and deleting **1000+** employee records.
- Implemented CRUD workflows connecting the frontend, REST API, and MongoDB.

Technical Skills

Programming: Python, SQL, Java, C++, JavaScript
Data Science: Pandas, NumPy, feature engineering, model evaluation, time-series analysis, and data visualization
Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, NLP, Computer Vision, OpenCV, LangChain
Cloud: AWS, Docker, Flask, REST APIs, CI/CD
Tools and Databases: Git, Jupyter, Streamlit, Tableau, Matplotlib, Seaborn, GeoPandas, MySQL, MongoDB

Certifications & Honors

AWS ML Foundations | AWS ML for NLP | Google Cloud Foundations: Data/ML/AI, Infrastructure, and Data Analytics
International Math Olympiad: 3x Gold, 1x Silver, **Top 100** State Rank